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Skipped uplink configured grant occasions in sidelink transmissions

Abstract

Aspects presented herein may enable a UE with an UL-CG to use resources associated with one or more UL transmissions for sidelink communications. In one aspect, a first UE transmits an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, where the indication is transmitted to at least one second UE. The first UE skips at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources. The first UE transmits a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to communication systems, and more particularly, to a wireless communication involving sidelink transmissions.

INTRODUCTION

(2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

(3) These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a

municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

(4) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus transmits an indication of an uplink (UL)-configured grant (CG) (UL-CG) configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first user equipment (UE) to a network node, where the indication is transmitted to at least one second UE. The apparatus skips at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources. The apparatus transmits a sidelink (SL) transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration.

(6) In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus receives an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node. The apparatus receives a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration.

(7) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating an example of a wireless communications system and an access network.

(2) FIG. **2**A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.

(3) FIG. **2**B is a diagram illustrating an example of downlink (DL) channels within a subframe, in accordance with various aspects of the present disclosure.

(4) FIG. **2**C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.

(5) FIG. **2**D is a diagram illustrating an example of uplink (UL) channels within a subframe, in

accordance with various aspects of the present disclosure.
(6) FIG. **3** is a diagram illustrating an example of a base station and user equipment (UE) in an access network.
(7) FIG. **4** is a diagram illustrating an example of wireless communication between wireless devices based on sidelink (SL) communication in accordance with various aspects of the present disclosure.
(8) FIG. **5**A is a diagram illustrating an example of a first resource allocation mode for SL communication in accordance with various aspects of the present disclosure.
(9) FIG. **5**B is a diagram illustrating an example of a second resource allocation mode for SL communication in accordance with various aspects of the present disclosure.
(10) FIG. **6** is a diagram illustrating an example structure of a sidelink resource pool in accordance with various aspects of the present disclosure.
(11) FIG. **7** is a diagram illustrating an example of sidelink resource reservations in accordance with various aspects of the present disclosure.
(12) FIG. **8** is a diagram illustrating an example resource selection timeline in accordance with various aspects of the present disclosure.
(13) FIG. **9** is a diagram illustrating an example configured grant (CG) configuration in accordance with various aspects of the present disclosure.
(14) FIG. **10** is a communication flow illustrating an example of a UE using a skipped UL transmission for sidelink communication in accordance with various aspects of the present disclosure.
(15) FIG. **11** is a flowchart of a method of wireless communication.
(16) FIG. **12** is a flowchart of a method of wireless communication.
(17) FIG. **13** is a diagram illustrating an example of a hardware implementation for an example apparatus and/or network entity.
(18) FIG. **14** is a flowchart of a method of wireless communication.
(19) FIG. **15** is a flowchart of a method of wireless communication.
(20) FIG. **16** is a diagram illustrating an example of a hardware implementation for an example apparatus and/or network entity.

DETAILED DESCRIPTION

(21) Aspects presented herein may improve resource utilization for a communication network. Aspects presented herein may enable a user equipment (UE) with an uplink (UL)-configured grant (CG) (UL-CG) to use resources associated with one or more UL transmissions for sidelink communications. For example, if a UE with an UL-CG determines to skip an UL transmission associated with the UL-CG (e.g., when the UL buffer is empty for the UE), the UE may use the resource associated with the skipped UL transmission for communicating with another UE over the sidelink (e.g., for transmitting a sidelink message to that UE or for receiving a sidelink massage from that UE, etc.). In addition, if other UEs in the sidelink are aware of the UE's UL-CG, they may be configured to monitor the resources associated with the UL-CG for sidelink transmission from the UE, which may reduce power consumption at these UEs.
(22) The detailed description set forth below in connection with the drawings describes various configurations and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.
(23) Several aspects of telecommunication systems are presented with reference to various apparatus and methods. These apparatus and methods are described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be

implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(24) By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a "processing system" that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise, shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.

(25) Accordingly, in one or more example aspects, implementations, and/or use cases, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

(26) While aspects, implementations, and/or use cases are described in this application by illustration to some examples, additional or different aspects, implementations and/or use cases may come about in many different arrangements and scenarios. Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects, implementations, and/or use cases may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described examples may occur. Aspects, implementations, and/or use cases may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques herein. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.

(27) Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core

network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

(28) An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

(29) Base station operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

(30) FIG. **1** is a diagram **100** illustrating an example of a wireless communications system and an access network. The illustrated wireless communications system includes a disaggregated base station architecture. The disaggregated base station architecture may include one or more CUs **110** that can communicate directly with a core network **120** via a backhaul link, or indirectly with the core network **120** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **125** via an E2 link, or a Non-Real Time (Non-RT) RIC **115** associated with a Service Management and Orchestration (SMO) Framework **105**, or both). A CU **110** may communicate with one or more DUs **130** via respective midhaul links, such as an F1 interface. The DUs **130** may communicate with one or more RUs **140** via respective fronthaul links. The RUs **140** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **140**.

(31) Each of the units, i.e., the CUs **110**, the DUs **130**, the RUs **140**, as well as the Near-RT RICs **125**, the Non-RT RICs **115**, and the SMO Framework **105**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or to transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or to transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), configured to receive or to transmit signals, or both, over a wireless transmission medium to one or more of the other units.

(32) In some aspects, the CU **110** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol

(PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **110**. The CU **110** may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **110** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as an E1 interface when implemented in an O-RAN configuration. The CU **110** can be implemented to communicate with the DU **130**, as necessary, for network control and signaling.

(33) The DU **130** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **140**. In some aspects, the DU **130** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation, demodulation, or the like) depending, at least in part, on a functional split, such as those defined by 3GPP. In some aspects, the DU **130** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **130**, or with the control functions hosted by the CU **110**.

(34) Lower-layer functionality can be implemented by one or more RUs **140**. In some deployments, an RU **140**, controlled by a DU **130**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **140** can be implemented to handle over the air (OTA) communication with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **140** can be controlled by the corresponding DU **130**. In some scenarios, this configuration can enable the DU(s) **130** and the CU **110** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

(35) The SMO Framework **105** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **105** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements that may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **105** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **190**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **110**, DUs **130**, RUs **140** and Near-RT RICs **125**. In some implementations, the SMO Framework **105** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **111**, via an O1 interface. Additionally, in some implementations, the SMO Framework **105** can communicate directly with one or more RUs **140** via an O1 interface. The SMO Framework **105** also may include a Non-RT RIC **115** configured to support functionality of the SMO Framework **105**.

(36) The Non-RT RIC **115** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence (AI)/machine learning (ML) (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **125**. The Non-RT RIC **115** may be coupled to or communicate with (such as via an AI interface) the Near-RT RIC **125**. The Near-RT RIC **125** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **110**, one or more DUs **130**, or both, as well as an O-eNB,

with the Near-RT RIC **125**.

(37) In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **125**, the Non-RT RIC **115** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **125** and may be received at the SMO Framework **105** or the Non-RT RIC **115** from non-network data sources or from network functions. In some examples, the Non-RT RIC **115** or the Near-RT RIC **125** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **115** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **105** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

(38) At least one of the CU **110**, the DU **130**, and the RU **140** may be referred to as a base station **102**. Accordingly, a base station **102** may include one or more of the CU **110**, the DU **130**, and the RU **140** (each component indicated with dotted lines to signify that each component may or may not be included in the base station **102**). The base station **102** provides an access point to the core network **120** for a UE **104**. The base stations **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The small cells include femtocells, picocells, and microcells. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links between the RUs **140** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to an RU **140** and/or downlink (DL) (also referred to as forward link) transmissions from an RU **140** to a UE **104**. The communication links may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

(39) Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL wireless wide area network (WWAN) spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, Bluetooth, Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.

(40) The wireless communications system may further include a Wi-Fi AP **150** in communication with UEs **104** (also referred to as Wi-Fi stations (STAs)) via communication link **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the UEs **104**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

(41) The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a "sub-6 GHz" band in various documents and articles. A similar nomenclature issue sometimes

occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

(42) The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHz), FR4 (71 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

(43) With the above aspects in mind, unless specifically stated otherwise, the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.

(44) The base station **102** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate beamforming. The base station **102** may transmit a beamformed signal **182** to the UE **104** in one or more transmit directions. The UE **104** may receive the beamformed signal from the base station **102** in one or more receive directions. The UE **104** may also transmit a beamformed signal **184** to the base station **102** in one or more transmit directions. The base station **102** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **102**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **102**/UE **104**. The transmit and receive directions for the base station **102** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

(45) The base station **102** may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), network node, network entity, network equipment, or some other suitable terminology. The base station **102** can be implemented as an integrated access and backhaul (IAB) node, a relay node, a sidelink node, an aggregated (monolithic) base station with a baseband unit (BBU) (including a CU and a DU) and an RU, or as a disaggregated base station including one or more of a CU, a DU, and/or an RU. The set of base stations, which may include disaggregated base stations and/or aggregated base stations, may be referred to as next generation (NG) RAN (NG-RAN).

(46) The core network **120** may include an Access and Mobility Management Function (AMF) **161**, a Session Management Function (SMF) **162**, a User Plane Function (UPF) **163**, a Unified Data Management (UDM) **164**, one or more location servers **168**, and other functional entities. The AMF **161** is the control node that processes the signaling between the UEs **104** and the core network **120**. The AMF **161** supports registration management, connection management, mobility management, and other functions. The SMF **162** supports session management and other functions. The UPF **163** supports packet routing, packet forwarding, and other functions. The UDM **164** supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The one or more location servers **168** are illustrated as including a Gateway Mobile Location Center (GMLC) **165** and a Location Management Function (LMF) **166**. However, generally, the one or more location servers **168** may include one or more location/positioning servers, which may include one or more of the GMLC

165, the LMF **166**, a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like. The GMLC **165** and the LMF **166** support UE location services. The GMLC **165** provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The LMF **166** receives measurements and assistance information from the NG-RAN and the UE **104** via the AMF **161** to compute the position of the UE **104**. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UE **104**. Positioning the UE **104** may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UE **104** and/or the serving base station **102**. The signals measured may be based on one or more of a satellite positioning system (SPS) **170** (e.g., one or more of a Global Navigation Satellite System (GNSS), global position system (GPS), non-terrestrial network (NTN), or other satellite position/location system), LTE signals, wireless local area network (WLAN) signals, Bluetooth signals, a terrestrial beacon system (TBS), sensor-based information (e.g., barometric pressure sensor, motion sensor), NR enhanced cell ID (NR E-CID) methods, NR signals (e.g., multi-round trip time (Multi-RTT), DL angle-of-departure (DL-AoD), DL time difference of arrival (DL-TDOA), UL time difference of arrival (UL-TDOA), and UL angle-of-arrival (UL-AoA) positioning), and/or other systems/signals/sensors.

(47) Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.

(48) Referring again to FIG. **1**, in certain aspects, the UE **104** may be configured to transmit an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, where the indication is transmitted to at least one second UE; skip at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources; and transmit a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration (e.g., via the UL-CG/SL Tx configuration component **198**).

(49) In certain aspects, the UE **104** may be configured to receive an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node; and receive a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration (e.g., via the SL process component **197**).

(50) FIG. **2**A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. **2**B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. **2**C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. **2**D is a diagram **280** illustrating an example of UL channels within a 5G NR

subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. **2**A, **2**C, the 5G NR frame structure is assumed to be TDD, with subframe 4 being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe 3 being configured with slot format 1 (with all UL). While subframes 3, 4 are shown with slot formats 1, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.

(51) FIGS. **2**A-**2**D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (also referred to as single carrier frequency-division multiple access (SC-FDMA) symbols) (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) (see Table 1). The symbol length/duration may scale with 1/SCS.

(52) TABLE-US-00001 TABLE 1 Numerology, SCS, and CP SCS μ Δf = 2.sup.μ .Math. 15[KHz] Cyclic prefix 0 15 Normal 1 30 Normal 2 60 Normal, Extended 3 120 Normal 4 240 Normal 5 480 Normal 6 960 Normal

(53) For normal CP (14 symbols/slot), different numerologies μ 0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per subframe. Accordingly, for normal CP and numerology μ, there are 14 symbols/slot and 2.sup.μ slots/subframe. The subcarrier spacing may be equal to 2.sup.μ*15 kHz, where μ is the numerology 0 to 4. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=4 has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **2**A-**2**D provide an example of normal CP with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μs. Within a set of frames, there may be one or more different bandwidth parts (BWPs) (see FIG. **2**B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

(54) A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

(55) As illustrated in FIG. **2**A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS),

beam refinement RS (BRRS), and phase tracking RS (PT-RS).
(56) FIG. **2**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE **104** to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIB s), and paging messages.
(57) As illustrated in FIG. **2**C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.
(58) FIG. **2**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.
(59) FIG. **3** is a block diagram of a base station **310** in communication with a UE **350** in an access network. In the DL, Internet protocol (IP) packets may be provided to a controller/processor **375**. The controller/processor **375** implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor **375** provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIB s), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity

verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.
(60) The transmit (TX) processor **316** and the receive (RX) processor **370** implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor **316** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator **374** may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **350**. Each spatial stream may then be provided to a different antenna **320** via a separate transmitter **318**Tx. Each transmitter **318**Tx may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.
(61) At the UE **350**, each receiver **354**Rx receives a signal through its respective antenna **352**. Each receiver **354**Rx recovers information modulated onto an RF carrier and provides the information to the receive (RX) processor **356**. The TX processor **368** and the RX processor **356** implement layer 1 functionality associated with various signal processing functions. The RX processor **356** may perform spatial processing on the information to recover any spatial streams destined for the UE **350**. If multiple spatial streams are destined for the UE **350**, they may be combined by the RX processor **356** into a single OFDM symbol stream. The RX processor **356** then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **310**. These soft decisions may be based on channel estimates computed by the channel estimator **358**. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station **310** on the physical channel. The data and control signals are then provided to the controller/processor **359**, which implements layer 3 and layer 2 functionality.
(62) The controller/processor **359** can be associated with a memory **360** that stores program codes and data. The memory **360** may be referred to as a computer-readable medium. In the UL, the controller/processor **359** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets. The controller/processor **359** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.
(63) Similar to the functionality described in connection with the DL transmission by the base station **310**, the controller/processor **359** provides RRC layer functionality associated with system information (e.g., MIB, SIB s) acquisition, RRC connections, and measurement reporting; PDCP

layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.
(64) Channel estimates derived by a channel estimator **358** from a reference signal or feedback transmitted by the base station **310** may be used by the TX processor **368** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor **368** may be provided to different antenna **352** via separate transmitters **354**Tx. Each transmitter **354**Tx may modulate an RF carrier with a respective spatial stream for transmission.
(65) The UL transmission is processed at the base station **310** in a manner similar to that described in connection with the receiver function at the UE **350**. Each receiver **318**Rx receives a signal through its respective antenna **320**. Each receiver **318**Rx recovers information modulated onto an RF carrier and provides the information to a RX processor **370**.
(66) The controller/processor **375** can be associated with a memory **376** that stores program codes and data. The memory **376** may be referred to as a computer-readable medium. In the UL, the controller/processor **375** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets. The controller/processor **375** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.
(67) At least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359** may be configured to perform aspects in connection with the UL-CG/SL Tx configuration component **198** of FIG. **1**.
(68) FIG. **4** is a diagram **400** illustrating an example of wireless communication between wireless devices based on sidelink (SL) communication in accordance with various aspects of the present disclosure. In one example, a UE **402** may transmit a transmission **414**, e.g., including a control channel (e.g., a physical sidelink control channel (PSCCH)) and a corresponding data channel (e.g., a physical sidelink shared channel (PSSCH)), that may be received by one or more UEs (e.g., UEs **404** and **406**). A control channel may include information for decoding the corresponding data channel, and it may also be used by a receiving UE for avoiding interference (e.g., UEs **404** and **406** may be refrained from transmitting data on resources occupied/reserved by the UE **402**). For example, the UE **402** may indicate the number of transmission time intervals (TTIs) and the resource blocks (RBs) that are to be occupied by a transmission from the UE **402** in a control message (e.g., a sidelink control information (SCI) message). The UEs **402**, **404**, **406**, and **408** may each have the capability to operate as a transmitting UE in addition to operating as a receiving UE. For example, UEs **404**, **406**, and **408** may also transmit transmissions **422**, **416**, and **420**, respectively, to other UEs, such as the UEs **402** and **404**. The transmissions **414**, **416**, **420** may be broadcast or multicast to nearby wireless devices or UEs. For example, the UE **402** may transmit communication (e.g., data) for receipt by other UEs within a range **401** of the UE **402**. Additionally, or alternatively, a road side unit (RSU) **407** may be used to provide connectivity and information to sidelink devices, such as by receiving communication from and/or transmitting communication (e.g., communication **418**) to UEs **402**, **406**, and **408**.
(69) Sidelink communication that is exchanged directly between UEs (which may be referred to as “sidelink UEs” hereafter) may include discovery messages for a UE to find other nearby UEs. In some examples, the sidelink communication may also include resource reservation information associated with other sidelink UEs, which may be used by a UE for determining/selecting the

resources for transmission.
(70) In one example, a sidelink communication may be based on different types or modes of resource allocation mechanisms. As shown by a diagram **500**A of FIG. **5**A, in a first resource allocation mode (which may be referred to as “Mode 1,” “sidelink transmission Mode 1,” and/or “V2X Mode 1,” etc.), a centralized resource allocation may be provided. For example, under the first resource allocation mode, a base station **502** may determine and allocate sidelink resources for communications between a first UE **504** and a second UE **506**. The first UE **504** may receive an indication of the allocated sidelink resources (e.g., a resource grant) from the base station **502** via a UE-to-universal mobile telecommunications system (UMTS) terrestrial radio access network (UE-to-UTRAN) (Uu) link (e.g., via a resource radio control (RRC) message or downlink control information (DCI) (e.g., DCI format 3_0)), and then the first UE **504** may use the allocated sidelink resources for communicating with the second UE **506** over the sidelink (which may also be referred to as a PC5 link).
(71) As shown by a diagram **500**B of FIG. **5**B, in a second resource allocation mode (which may be referred to as “Mode 2,” “sidelink transmission Mode 2,” and/or “V2X Mode 2,” etc.), a distributed resource allocation may be provided between UEs. For example, under the second resource allocation mode, each UE may autonomously determine sidelink resources for its sidelink transmission. In order to coordinate the selection of sidelink resources by individual UEs, each UE may use a sensing technique to monitor/detect sidelink resources reserved/used by other UEs, and then each UE may select sidelink resources for its sidelink transmissions from unreserved/used sidelink resources. For example, a first UE **504** may sense and select unreserved/unused sidelink resources in a sidelink resource pool based on decoding SCI messages received (e.g., transmitted from a second UE **506** or another UE), and the first UE **504** may use the selected side resources for communicating with the second UE **506**. After the first UE **504** selects the sidelink resources for its transmission, the first UE **504** may also transmit/broadcast (e.g., via groupcast or broadcast) to other UEs the sidelink resources used/reserved by the first UE **504** via SCI, such that other UEs may refrain using these sidelink resources to avoid resource collision (e.g., two UEs transmitting simultaneously using same time and frequency resources). Signaling on sidelink may be the same between the two resource allocation modes (e.g., Mode 1 and Mode 2). For example, from a receiving UE's point of view (e.g., the second UE **506**), there may be no difference between the two resource allocation modes.
(72) In some examples, a UE receiving a sidelink transmission (which may be referred to as a receiving UE) may be configured to provide feedback (e.g., an acknowledgement (ACK) or a negative acknowledgement (NACK)) to a UE transmitting the sidelink transmission (which may be referred to as a transmitting UE). For example, after the second UE **506** receives a transmission from the first UE **504**, the second UE **506** may send an ACK to the first UE **504** via a physical sidelink feedback channel (PSFCH) if the second UE **506** is able to receive and decode the transmission. On the other hand, if the second UE **506** is unable to decode or receive the transmission, the second UE **506** may send a NACK to the first UE **504**. In one example, if the transmission from the first UE **504** is a unicast or a groupcast message, the second UE **506** may be configured to transmit an explicit ACK/NACK to the first UE **504** indicating whether the transmission is successfully decoded, e.g., the second UE **506** transmits an ACK if the transmission is successfully decoded and transmits a NACK if the transmission is not successfully decoded. In another example, if the transmission from the first UE **504** is a groupcast message, the second UE **506** may be configured to transmit an implicit NACK, where the second UE **506** may transmit a NACK to the first UE **504** if the second UE **506** is unable to decode or does not receive the transmission. However, the second UE **506** may skip transmitting an ACK if the second UE **506** successfully decodes the transmission.
(73) Sidelink communications may take place in transmission or reception sidelink resource pools. The minimum resource allocation unit in a sidelink resource pool may be a sub-channel in

frequency, and the resource allocation in time may be one slot. Some slots in a sidelink resource pool may are not be available for sidelink communications (e.g., may be reserved/configured for other purposes or other types of communications). For example, some slots may contain feedback resources (e.g., a PSFCH). A base station may configure a sidelink resource pool to a set of UEs, such as via an RRC configuration, or the sidelink resource pool may be preloaded on the set of UEs (e.g., via a pre-configuration).

(74) FIG. **6** is a diagram **600** illustrating an example structure of a sidelink resource pool in accordance with various aspects of the present disclosure. A sidelink resource pool **602** may include a set of time and frequency resources (e.g., each slot and sub-channel may indicate a time and frequency resource), and each time and frequency resource may be used by a transmitting UE for transmitting a PSCCH and/or a PSSCH, or used by a receiving UE for transmitting a PSFCH. For example, as shown at **604**, a slot **606** may include resources for a PSCCH **608**, a PSSCH **610**, and a PSFCH **612**. After a receiving UE receives the sidelink slot **606** (e.g., the second UE **506**), the receiving UE may first decode the SCI in the PSCCH **608** and/or the PSSCH **610** (e.g., for a two-stage SCI), then decode data in the PSSCH **610**. The receiving UE may also receive a feedback via the PSFCH **612** (e.g., for a previous transmission to the transmitting UE), or the receiving UE may also transmit a feedback for the PSSCH **610** via the PSFCH **612**.

(75) FIG. **7** is a diagram **700** illustrating an example of sidelink resource reservations in accordance with various aspects of the present disclosure. A transmitting UE (e.g., the first UE **504**) may determine whether a sidelink resource is available by decoding SCI(s) it receives from other UEs, and the transmitting UE may measure reference signal received power (RSRP) for reservations in decoded SCI(s). The RSRP of the transmission associated with SCI reserving resources may be projected onto a resource selection window. The transmitting UE may measure the RSRP on PSCCH or PSSCH DMRS according to a (pre-)configuration. The length of the sensing window (e.g., where SCI is decoded) may also be (pre-)configured. In addition, each reservation may include a priority indicated in SCI that is also tracked as part of sensing.

(76) Sidelink resource(s) reserved for a transmitting UE (e.g., the first UE **504**) may include one or more sub-channels (SCs) in the frequency domain (e.g., SC1 to SC 4) and a slot in the time domain. For example, as shown at **702**, a first transmitting UE and a second transmitting UE may use resources in a current slot (e.g., slot 1) for transmission, and/or for sensing and reserving resources in future slots (e.g., for transmission/retransmissions). In some examples, as shown at **704**, a UE may reserve up to two future resources and the resource reservation may be limited to a resource selection window with defined slots and sub-channels. For example, the first transmitting UE (e.g., UE 1) may reserve two future sidelink resources **706** and **708** in the resource selection window, and the second transmitting UE (e.g., UE 2) may reserve two future sidelink resources **710** and **712** in the resource selection window, etc. After selecting and/or reserving the sidelink resources for transmission/retransmission, the first transmitting UE and the second transmitting UE may transmit its resource reservation information to other UEs via SCI.

(77) A sidelink resource reservation may be periodic or aperiodic. If the resource reservation is configured to be periodic, the periodic resource reservation may be turned on or off by a configuration in a sidelink resource pool. Also, a UE may be configured to continue monitoring resource reservation messages (e.g., SCIs) sent by other UE(s) or stations so that the UE may maintain a sensing history regarding which resources are being used and/or reserved. Then, the UE may perform resource selection based at least in part on the sensing history when the UE has a transmission. The UE may maintain reservation information for a period of time, e.g., within a sensing window (e.g., as shown at **702**). The length of the sensing window may be configured for the UE, such as by a base station. Each resource reservation may have a priority level indicated in the SCI, such that a transmission with a higher priority reservation may pre-empt a transmission with a lower priority reservation.

(78) As described above, a transmitting UE may perform signal/channel measurement for a sidelink

resource that has been reserved and/or used by other UE(s), such as by measuring the RSRP of the SCI that reserves the sidelink resource. Based at least in part on the signal/channel measurement, the transmitting UE may consider using/reusing the sidelink resource that has been reserved by other UE(s). For example, a UE may consider a reserved resource to be available when the RSRP measured for the SCI reserving the resource is below an RSRP threshold, and the UE may use/reuse that reserved resource for transmission. When the measured RSRP is below the RSRP threshold, it may indicate that the UE reserving the resource may be distant or not using the reserved resource. As such, the use/reuse of the reserved resource may be less likely to cause interference or impact to that UE. In some examples, the RSRP threshold may be determined or changed based on the amount of available resources in a resource selection window. For example, if the amount of available resources is below a threshold/percentage (e.g., below 20%) within a resource selection window, the UE may be configured to use an increased/higher RSRP threshold so that the UE is more likely to be able to reuse resources reserved by other UEs. The RSRP comparison threshold may be (pre-)configured per transmitter priority and receiver priority pair. In addition, a sidelink (or packet) transmission or retransmission may be configured with a packet delay budget (PDB), which may provide a time in which the sidelink transmission or retransmission is to be transmitted by a UE. If the UE is unable to transmit the sidelink transmission within the PDB, the UE may be configured to abort or discard the transmission, and restart the sidelink resource sensing and selection process.
(79) FIG. **8** is a diagram **800** illustrating an example resource selection timeline in accordance with various aspects of the present disclosure. A UE may select sidelink resources from a resource selection window **802**, which is illustrated as having sixteen (16) resource blocks formed by two sub-channels and eight slots in this example. The duration of the resource selection window **802** may be represented by [n+T.sub.1, n+T.sub.2], where n may represent the time a resource selection is triggered, such as shown at **804** when the UE is selecting a sidelink resource for transmission. T.sub.1 may represent the starting of the resource selection window **802** and T.sub.2 may represent the end of the resource selection window **802**. The length of the resource selection window **802** may vary depending on the configuration, and may be configured for the UE via an upper layer (e.g., from 20 ms to 100 ms etc.). In some examples, the value of T.sub.0 may be (pre-)configured to be 100 ms or 1100 ms. The value of T.sub.1 may be up to the UE implementation (e.g., selected from a time between zero (0) and the time it takes the UE to process and implement the resource selection (e.g., T.sub.proc,1)), and the value of T.sub.2 may be selected from a time between T.sub.2,min and the remaining of a PDB duration associated with the transmission. In some examples, T.sub.2,min may be configured per each sidelink priority value which may be indicated in SCI from the following set of values: {1, 5, 10, 20}.Math.2.sup.μ-1 slots, where μ=1, 2, 3, 4 for 15, 30, 60, 120 kHz sub-carrier spacing (SCS), respectively. PDB may be used to define the upper limit of a delay suffered by a packet between the UE and policy and charging enforcement function (PCEF). A UE may be specified to confirm resource availability at least time T.sub.3 before transmission.
(80) For example, as a UE may sense and decode sidelink messages (e.g., SCIs) from other UE(s), a sensing history (or resource use/reservation history) may be recorded/determined from a sensing window **806** and used by the UE to determine which resources in the resource selection window **802** are reserved by other UE(s). For example, the sensing window **806** may represent a time duration prior to the resource selection during which the UE may collect resource reservation related information from other UEs (e.g., via their SCIs). In one example, the time duration may be the last 1,000 ms prior to the resource selection triggering at **804**. In other examples, the sensing window **806** may have a duration between 100 ms to 1,100 ms, as represented by T.sub.0 in FIG. **8**, minus the time it takes the UE to process the sensing window **806** (e.g., T.sub.proc,0).
(81) Based on the resource reservation related information obtained during the sensing window **806**, the UE may determine available candidate resources for the resource selection window **802**.

For example, based on the resource reservation information decoded from SCIs of other UEs (e.g., UE 1, UE 2, and UE 3), the UE may discover that UE1 has transmitted a data during the sensing window **806** and has reserved resource blocks **808** and **810** in the resource selection window **802**. Similarly, based on the sensing window **806**, the UE may also discover that resource blocks **812** and **814** have been reserved by UE2 and UE3, respectively. As such, the UE may determine that resource blocks **808**, **810**, **812** and **814** in the resource selection window **802** are not available for selection and may excluded these resource blocks from selection. Then, the UE may identify the remaining resource blocks as available candidate resources. The UE may perform the sensing up to a defined duration (e.g., T.sub.3) before its transmission.
(82) FIG. **9** is a diagram **900** illustrating an example configured grant (CG) configuration in accordance with various aspects of the present disclosure. In some scenarios, a network entity (e.g., a base station) may schedule a set of uplink (UL) transmissions for a UE using a CG (which may be referred to as an UL-CG), such that the base station may avoid requesting and assigning resources for each UL transmission from the UE by pre-allocating resources to the UE. For example, a base station may configure a UE with an UL-CG via an UL-CG configuration, where the UL-CG configuration may include a set of parameters (e.g., resource allocation, periodicities of resources, frequency hopping type, MCS used, etc., as shown by FIG. **9**) that configures the UE with a set of UL transmissions, such that the UE may transmit UL transmissions using specified time and frequency resources at a defined periodicity. As such, an UL-CG may be defined by time and frequency resources and similarly periodicity, repetition, and other configurations. In one example, an UL-CG may have two types, where one type may not specify a DCI activation whereas the other type may specify a DCI activation (e.g., the CG is activated for the UE via a DCI message). In some examples, if a UE does not have anything to transmit in an UL transmission occasion (e.g., an UL transmission in the set of UL transmissions configured by the UL-CG), the UE may skip transmission in that UL transmission occasion (which may be referred to as "UL-CG skipping"). In other words, UL-CG skipping is a feature that enables a UE to skip the transmission of CG in case the UL buffer is empty. For purposes of the present disclosure, if a wireless device, such as a UE, skips a transmission, the wireless device may be configured not to transmit anything for that transmission or transmit zero data for that transmission.
(83) While a UE may use the UL-CG skipping feature to skip one or more UL transmissions configured based on the CG (which may be referred to as "UL-CG instance(s)" or "UL transmission instance(s)"), resources associated with the skipped UL transmission(s) may be wasted. As resources associated with a CG may be reserved for a specified UE, other UE(s) may not be able to use it because they may not know whether the UE is going to skip any of the UL transmissions.
(84) Aspects presented herein may improve resource utilization for a communication network. Aspects presented herein may enable a UE with an UL-CG to use resources associated with one or more UL transmissions for sidelink communications. For example, if a UE with an UL-CG determines to skip an UL transmission associated with the UL-CG (e.g., when the UL buffer is empty for the UE), the UE may use the resource associated with the skipped UL transmission for communicating with another UE over the sidelink (e.g., for transmitting a sidelink message to that UE or for receiving a sidelink massage from that UE, etc.). In addition, if other UEs in the sidelink are aware of the UE's UL-CG, they may be configured to monitor the resources associated with the UL-CG for sidelink transmission from the UE, which may reduce power consumption at these UEs. In some examples, the UL-CG and the sidelink resources may be within the same carrier.
(85) FIG. **10** is a communication flow **1000** illustrating an example of a UE using a skipped UL transmission for sidelink communication in accordance with various aspects of the present disclosure. The numberings associated with the communication flow **1000** do not specify a particular temporal order and are merely used as references for the communication flow **1000**. Aspects presented herein may enable a network entity (e.g., a base station) and/or a transmitting

UE (e.g., a UE with an UL-CG) to share the transmitting UE's UL-CG configurations with a subset of sidelink UEs (e.g., receiving UEs), where the network entity may share this configuration over a Uu link and the UE may share this configuration over sidelink. The shared UL-CG configurations may not include the entire set of parameters of the UL-CG configurations but a subset of the parameters, such as just the time and frequency resources as well as the periodicity for the UL transmissions. In response, a receiving UE may be configured to perform reception in all UL-CG occasions of the transmitting UE, perform reception in an RRC configured subset of UL-CG occasions of the transmitting UE, or perform reception when it receives an indication from the transmitting UE. In some examples, the transmitting UE may be configured with an allowance to drop the UL transmission to use it for sidelink. The transmitting UE may also skip/free-up multiple UL-CG occasions and use it for sidelink depending on the priority order between the UL and sidelink.
(86) At **1020**, a base station **1002** (or a network node) may transmit an UL-CG configuration **1008** to a first UE **1004**, where the UL-CG configuration **1008** may include a set of periodic resources **1010** in which the first UE **1004** may use for transmitting UL transmissions to the base station **1002**, such as described in connection with FIG. **9**. For example, the set of periodic resources **1010** may include a first CG resource (CG 1), a second CG resource (CG 2), a third CG resource (CG 3), and up to N.sup.th CG resource (CG N). UL transmission based on each CG resource may be referred to as an UL-CG occasion. For purposes of the present disclosure, a periodic resource may refer to a time and frequency resource that is available at a specified interval (e.g., every X slots, every Y ms, etc.) for transmission and/or reception. For example, a periodic resource may be a time and frequency resource (e.g., symbol(s) and a bandwidth) that is available to a UE at every two slots, where the UE may use each periodic resource for transmitting UL data to a base station.
(87) At **1022**, the base station **1002** and/or the first UE **1004** may transmit the UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** to at least a second UE **1006** (or a subset of sidelink UEs). For example, the base station **1002** may transmit the UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** to the second UE **1006** over a Uu link, and the first UE **1004** may transmit the UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** to the second UE **1006** via a sidelink.
(88) Based on the UL-CG configuration **1008**, the second UE **1006** may know the set of periodic resources **1010** (e.g., CG **1** to CG N) that is configured for the first UE **1004**. In one example, as the second UE **1006** may be just specified (or interested) to know the set of periodic resources **1010** configured for the first UE **1004**, the base station **1002** and/or the first UE **1004** may transmit a portion of the UL-CG configuration **1008** to the second UE **1006** instead of the entire set of the UL-CG configuration **1008** (e.g., as shown by FIG. **9**) to reduce signaling overhead. For example, the base station **1002** and/or the first UE **1004** may just indicate the time, the frequency, and the periodicity (e.g., between two periodic resources) associated with the set of periodic resources **1010** to the second UE **1006**.
(89) At **1024**, the first UE **1004** may transmit UL transmissions to the base station **1002** based on the UL-CG configuration **1008**, such as using the set of periodic resources **1010** for transmitting data to the base station **1002**.
(90) At **1026**, in response to the UL-CG configuration **1008** or the portion of the UL-CG configuration **1008** received from the base station **1002** and/or the first UE **1004**, the second UE **1006** may monitor for one or more SL transmissions from the first UE **1004** (e.g., based on the set of periodic resources **1010**).
(91) In one example, the second UE **1006** may be configured to monitor sidelink transmissions from the first UE **1004** based on all UL-CG occasions (e.g., CG 1 to CG N) associated with the set of periodic resources **1010**. In another example, to reduce power consumption at the second UE **1006**, the second UE **1006** may be configured to monitor sidelink transmissions from the first UE **1004** based on a subset of UL-CG occasions associated with the set of periodic resources **1010**. For

example, the second UE **1006** may be configured to monitor sidelink transmissions on odd UL-CG occasions (e.g., on CG 1, CG 3, CG 5, etc.), on even UL-CG occasions (e.g., on CG 2, CG 4, CG 6, etc.), or on every X UL-CG occasion (e.g., on CG 3, CG 6, CG 9, etc. for X=3). The base station **1002** may configure the second UE **1006** to monitor the subset of UL-CG occasions via an RRC configuration. In another example, to further reduce power consumption at the second UE **1006**, the second UE **1006** may be configured to monitor for sidelink transmissions from the first UE **1004** after the second UE receives an indication from the first UE **1004**, such as shown at **1028** (described in more details below).

(92) At **1030**, the first UE **1004** may determine to skip the at least one UL transmission associated with the UL-CG configuration **1008**. For example, the first UE **1004** may not have anything to transmit for two subsequent UL-CG occasions (e.g., CG 4 and CG 5). If the first UE **1004** is communicating with the second UE **1006** over sidelink, the first UE **1004** may use at least a portion of resources for these two subsequent UL-CG occasions for the sidelink communication.

(93) For example, at **1032**, after determining to skip two subsequent UL-CG occasions (e.g., CG 4 and CG 5), the first UE **1004** may transmit sidelink transmission(s) to the second UE **1006** using resources (or a portion of the resources) configured for these two subsequent UL-CG occasions (e.g., resources for CG 4 and CG 5). In another example, the first UE **1004** may also receive sidelink transmission(s) from the second UE **1006** via these resources if the second UE **1006** is aware that the first UE **1004** is skipping these two UL-CG occasions (e.g., based on an indication from the first UE **1004**).

(94) In some examples, as described above, the second UE **1006** may be configured to monitor for sidelink transmissions from the first UE **1004** based on the indication from the first UE **1004**. As such, at **1028**, if the first UE **1004** determines to use the skipped UL-CG resources for communicating with the second UE **1006**, the first UE **1004** may transmit an indication **1012** to inform the second UE **1006** regarding the sidelink transmission.

(95) In another example, the first UE **1004** may skip one or more UL transmission based on the base station **1002**'s permission. For example, the base station **1002** may transmit a configuration for an allowance to skip at least one UL transmission (or to skip up to X UL transmissions) to the first UE **1004**, and the first UE **1004** may skip at least one UL transmission (or to skip up to X UL transmissions) based on the configuration.

(96) In another example, the first UE **1004** may determine to skip one or more UL transmissions based on a priority order between an UL transmission and a sidelink transmission. For example, if a sidelink transmission is associated with a higher priority compared to an UL transmission, the first UE **1004** may skip/drop the UL transmission and use the skipped/dropped resources for the SL transmission. Under such example, the first UE **1004** may skip/drop the UL transmission even if there are data (e.g., Uu data) scheduled to be transmitted using the UL transmission. In addition, the first UE **1004** may also skip more than one UL transmission if the sidelink transmission (with higher priority) specifies additional transmission resources.

(97) In another example, the first UE **1004** may determine whether to skip an UL transmission (or whether it is permitted to skip an UL transmission) based on the slot type configured for the UL transmission. For example, a slot configured for the first UE **1004** may include an UL slot, a flexible (FL) slot, a full-duplex (FD) slot, and/or a half-duplex (HD) slot. Thus, the first UE **1004** may be permitted to skip an UL transmission is the UL transmission is configured with an UL slot, and the first UE **1004** may not be permitted to skip an UL transmission is the UL transmission is configured with an FD slot, etc. A half-duplex transmission may refer to a one-way transmission between two UEs (e.g., a transmitting UE and a receiving UE), whereas a full-duplex may refer to a two-way transmission between two UEs at the same time.

(98) In another example, prior to **1032**, the first UE **1004** may receive a configuration (e.g., from the base station **1002**) that configures one or more parameters for transmitting the SL transmission, where the one or more parameters may be based on at least one configuration parameter of the UL-

CG configuration **1008**. For example, as the base station **1002** knows the parameters associated with the UL-CG configuration **1008** (e.g., the time and frequency resource, the beam direction, the periodicity, the bandwidth, the SCS, etc.) for the first UE **1004**, the base station **1002** may determine the beam direction, the modulation coding scheme (MCS), the bandwidth, and/or the transmission (Tx) power for the first UE **1004** to transmit the SL transmission(s) based on some of these parameters. Similarly, the second UE **1006** may also receive a configuration (e.g., from the base station **1002**) that configures one or more parameters for receiving the SL transmission, where the one or more parameters may be based on at least one configuration parameter of the UL-CG configuration **1008**. For example, as the base station **1002** knows the parameters associated with the UL-CG configuration **1008** for the first UE **1004**, the base station **1002** may determine the beam direction, the MCS, the bandwidth, and/or the reception (Rx) power for the second UE **1006** to receive the SL transmission(s) based on some of these parameters.
(99) At **1034**, after the sidelink transmission is performed (e.g., by either the first UE **1004** or the second UE **1006**), if the first UE **1004** has data to be transmitted to the base station **1002**, the first UE **1004** may resume UL transmissions based on the UL-CG configuration **1008**.
(100) FIG. **11** is a flowchart **1100** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, **402**, **404**, **406**, **504**, **506**, **1004**; the apparatus **1304**). The method may enable the UE with an UL-CG to use resources associated with one or more UL transmissions for sidelink communications.
(101) At **1102**, the UE may receive a UL-CG configuration from a network node, where the UL-CG configuration includes a set of periodic resources, such as described in connection with FIG. **10**. For example, at **1020**, the first UE **1004** may receive an UL-CG configuration **1008** from the base station **1002**, where the UL-CG configuration **1008** includes a set of periodic resources **1010**. The reception of the UL-CG configuration may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.
(102) At **1104**, the UE may transmit an indication of the UL-CG configuration including the set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, and the indication is transmitted to at least one second UE, such as described in connection with FIG. **10**. For example, at **1022**, the first UE **1004** may transmit the UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** to the second UE **1006**, where the UL-CG configuration **1008** includes the set of periodic resources **1010**. The transmission of the indication of the UL-CG configuration may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.
(103) In one example, transmitting the indication of the UL-CG configuration may include forwarding the UL-CG configuration received from the network node.
(104) In another example, the indication includes a subset of parameters of the UL-CG configuration received from the network node.
(105) In another example, the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.
(106) In another example, the UE corresponds to a transmitting SL UE and the at least one second UE corresponds to a receiving SL UE.
(107) At **1106**, the UE may receive a configuration for an allowance to skip the at least one UL transmission from the network node, where at least one UL transmission is skipped based on the configuration for the allowance, such as described in connection with FIG. **10**. For example, the first UE **1004** may receive a configuration for an allowance to skip the at least one UL transmission from the base station **1002**, where at least one UL transmission is skipped based on the configuration for the allowance. The reception of the configuration for an allowance to skip the at least one UL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component

198, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.

(108) At **1108**, the UE may determine to skip the at least one UL transmission associated with the UL-CG configuration based on a slot type configured for the at least one UL transmission, where at least one UL transmission is skipped based on the determination, such as described in connection with FIG. **10**. For example, at **1030**, the first UE **1004** may determine to skip the at least one UL transmission associated with the UL-CG configuration **1008** based on a slot type configured for the at least one UL transmission, where at least one UL transmission is skipped based on the determination. The determination to skip the at least one UL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**. In one example, the slot type includes at least one of: an UL slot, a flexible slot, a full-duplex slot, or a half-duplex slot.

(109) At **1110**, the UE may receive a configuration that configures one or more parameters for transmitting the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration, such as described in connection with FIG. **10**. For example, the first UE **1004** may receive a configuration from the base station **1002** that configures one or more parameters for transmitting the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration **1008**. The reception of the configuration that configures one or more parameters for transmitting the SL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**. In one example, the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or a Tx power for transmitting the SL transmission.

(110) At **1112**, the UE may skip at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources, such as described in connection with FIG. **10**. For example, at **1030** and **1032**, the first UE **1004** may skip at least one UL transmission associated with the UL-CG configuration **1008**, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources **1010**. The skipping of at least one UL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.

(111) In one example, the at least one UL transmission is skipped based on a priority order between UL communication and SL communication. In such an example, a plurality of UL transmissions is skipped based on the SL transmission having a higher priority order than the plurality of UL transmissions.

(112) At **1114**, the UE may transmit a second indication to the at least one second UE indicating the SL transmission, where the SL transmission is transmitted based on the second indication, such as described in connection with FIG. **10**. For example, at **1028**, the first UE **1004** may transmit an indication **1012** to the second UE **1006** indicating the SL transmission, where the SL transmission is transmitted to the second UE **1006** based on the indication **1012**. The transmission of the second indication may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.

(113) At **1116**, the UE may transmit a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration, such as described in connection with FIG. **10**. For example, at **1032**, the first UE **1004** may transmit a SL transmission to the second UE **1006** using at least the subset of the set of periodic resources **1010** corresponding to the skipped at least one UL transmission associated with the UL-CG configuration **1008**. The transmission of the SL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.

(114) FIG. **12** is a flowchart **1200** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, **402**, **404**, **406**, **504**, **506**, **1004**; the apparatus **1304**). The method may enable the UE with an UL-CG to use resources associated with one or more UL transmissions for sidelink communications.
(115) At **1204**, the UE may transmit an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, and the indication is transmitted to at least one second UE, such as described in connection with FIG. **10**. For example, at **1022**, the first UE **1004** may transmit the UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** to the second UE **1006**, where the UL-CG configuration **1008** includes the set of periodic resources **1010**. The transmission of the indication of the UL-CG configuration may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.
(116) At **1212**, the UE may skip at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources, such as described in connection with FIG. **10**. For example, at **1030** and **1032**, the first UE **1004** may skip at least one UL transmission associated with the UL-CG configuration **1008**, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources **1010**. The skipping of at least one UL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.
(117) At **1216**, the UE may transmit a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration, such as described in connection with FIG. **10**. For example, at **1032**, the first UE **1004** may transmit a SL transmission to the second UE **1006** using at least the subset of the set of periodic resources **1010** corresponding to the skipped at least one UL transmission associated with the UL-CG configuration **1008**. The transmission of the SL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.
(118) In one example, the UE may receive the UL-CG configuration from a network node, where the UL-CG configuration includes a set of periodic resources, such as described in connection with FIG. **10**. For example, at **1020**, the first UE **1004** may receive an UL-CG configuration **1008** from the base station **1002**, where the UL-CG configuration **1008** includes a set of periodic resources **1010**. The reception of the UL-CG configuration may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.
(119) In another example, transmitting the indication of the UL-CG configuration may include forwarding the UL-CG configuration received from the network node.
(120) In another example, the indication includes a subset of parameters of the UL-CG configuration received from the network node.
(121) In another example, the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.
(122) In another example, the UE corresponds to a transmitting SL UE and the at least one second UE corresponds to a receiving SL UE.
(123) In another example, the UE may receive a configuration for an allowance to skip the at least one UL transmission from the network node, where at least one UL transmission is skipped based on the configuration for the allowance, such as described in connection with FIG. **10**. For example, the first UE **1004** may receive a configuration for an allowance to skip the at least one UL transmission from the base station **1002**, where at least one UL transmission is skipped based on the configuration for the allowance. The reception of the configuration for an allowance to skip the

at least one UL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.

(124) In another example, the UE may determine to skip the at least one UL transmission associated with the UL-CG configuration based on a slot type configured for the at least one UL transmission, where at least one UL transmission is skipped based on the determination, such as described in connection with FIG. **10**. For example, at **1030**, the first UE **1004** may determine to skip the at least one UL transmission associated with the UL-CG configuration **1008** based on a slot type configured for the at least one UL transmission, where at least one UL transmission is skipped based on the determination. The determination to skip the at least one UL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**. In one example, the slot type includes at least one of: an UL slot, a flexible slot, a full-duplex slot, or a half-duplex slot.

(125) In another example, the UE may receive a configuration that configures one or more parameters for transmitting the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration, such as described in connection with FIG. **10**. For example, the first UE **1004** may receive a configuration from the base station **1002** that configures one or more parameters for transmitting the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration **1008**. The reception of the configuration that configures one or more parameters for transmitting the SL transmission may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**. In one example, the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or a Tx power for transmitting the SL transmission.

(126) In another example, the at least one UL transmission is skipped based on a priority order between UL communication and SL communication. In such an example, a plurality of UL transmissions is skipped based on the SL transmission having a higher priority order than the plurality of UL transmissions.

(127) In another example, the UE may transmit a second indication to the at least one second UE indicating the SL transmission, where the SL transmission is transmitted based on the second indication, such as described in connection with FIG. **10**. For example, at **1028**, the first UE **1004** may transmit an indication **1012** to the second UE **1006** indicating the SL transmission, where the SL transmission is transmitted to the second UE **1006** based on the indication **1012**. The transmission of the second indication may be performed by, e.g., the UL-CG/SL Tx configuration component **198**, the cellular baseband processor **1324**, and/or the transceiver(s) **1322** of the apparatus **1304** in FIG. **13**.

(128) FIG. **13** is a diagram **1300** illustrating an example of a hardware implementation for an apparatus **1304**. The apparatus **1304** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **1304** may include a cellular baseband processor **1324** (also referred to as a modem) coupled to one or more transceivers **1322** (e.g., cellular RF transceiver). The cellular baseband processor **1324** may include on-chip memory **1324′**. In some aspects, the apparatus **1304** may further include one or more subscriber identity modules (SIM) cards **1320** and an application processor **1306** coupled to a secure digital (SD) card **1308** and a screen **1310**. The application processor **1306** may include on-chip memory **1306′**. In some aspects, the apparatus **1304** may further include a Bluetooth module **1312**, a WLAN module **1314**, an SPS module **1316** (e.g., GNSS module), one or more sensor modules **1318** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial measurement unit (IMU), gyroscope, and/or accelerometer(s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other

technologies used for positioning), additional memory modules **1326**, a power supply **1330**, and/or a camera **1332**. The Bluetooth module **1312**, the WLAN module **1314**, and the SPS module **1316** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **1312**, the WLAN module **1314**, and the SPS module **1316** may include their own dedicated antennas and/or utilize the antennas **1380** for communication. The cellular baseband processor **1324** communicates through the transceiver(s) **1322** via one or more antennas **1380** with the UE **104** and/or with an RU associated with a network entity **1302**. The cellular baseband processor **1324** and the application processor **1306** may each include a computer-readable medium/memory **1324′**, **1306′**, respectively. The additional memory modules **1326** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory **1324′**, **1306′**, **1326** may be non-transitory. The cellular baseband processor **1324** and the application processor **1306** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor **1324**/application processor **1306**, causes the cellular baseband processor **1324**/application processor **1306** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor **1324**/application processor **1306** when executing software. The cellular baseband processor **1324**/application processor **1306** may be a component of the UE **350** and may include the memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **1304** may be a processor chip (modem and/or application) and include just the cellular baseband processor **1324** and/or the application processor **1306**, and in another configuration, the apparatus **1304** may be the entire UE (e.g., see **350** of FIG. **3**) and include the additional modules of the apparatus **1304**.

(129) As discussed supra, the UL-CG/SL Tx configuration component **198** is configured to transmit an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, where the indication is transmitted to at least one second UE. The UL-CG/SL Tx configuration component **198** may also be configured to skip at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources. The UL-CG/SL Tx configuration component **198** may also be configured to transmit a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration. The UL-CG/SL Tx configuration component **198** may be within the cellular baseband processor **1324**, the application processor **1306**, or both the cellular baseband processor **1324** and the application processor **1306**. The UL-CG/SL Tx configuration component **198** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. As shown, the apparatus **1304** may include a variety of components configured for various functions. In one configuration, the apparatus **1304**, and in particular the cellular baseband processor **1324** and/or the application processor **1306**, includes means for transmitting an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, where the indication is transmitted to at least one second UE. The apparatus **1304** may further include means for skipping at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources. The apparatus **1304** may further include means for transmitting a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration.

(130) In one example, the apparatus **1304** may further include means for receiving the UL-CG configuration from a network node, where the UL-CG configuration includes a set of periodic resources.
(131) In another example, the means for transmitting the indication of the UL-CG configuration may include configuring the apparatus **1304** to forwarding the UL-CG configuration received from the network node.
(132) In another example, the indication includes a subset of parameters of the UL-CG configuration received from the network node.
(133) In another example, the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.
(134) In another example, the apparatus **1304** may further include means for receiving a configuration for an allowance to skip the at least one UL transmission from the network node, where at least one UL transmission is skipped based on the configuration for the allowance.
(135) In another example, the apparatus **1304** may further include means for determining to skip the at least one UL transmission associated with the UL-CG configuration based on a slot type configured for the at least one UL transmission, where at least one UL transmission is skipped based on the determination. In such an example, the slot type includes at least one of: an UL slot, a flexible slot, a full-duplex slot, or a half-duplex slot.
(136) In another example, the apparatus **1304** may further include means for receiving a configuration that configures one or more parameters for transmitting the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration. In such an example, the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or a Tx power for transmitting the SL transmission.
(137) In another example, the at least one UL transmission is skipped based on a priority order between UL communication and SL communication. In such an example, a plurality of UL transmissions is skipped based on the SL transmission having a higher priority order than the plurality of UL transmissions.
(138) In another example, the apparatus **1304** may further include means for transmitting a second indication to the at least one second UE indicating the SL transmission, where the SL transmission is transmitted based on the second indication.
(139) The means may be the UL-CG/SL Tx configuration component **198** of the apparatus **1304** configured to perform the functions recited by the means. As described supra, the apparatus **1304** may include the TX processor **368**, the RX processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX processor **368**, the RX processor **356**, and/or the controller/processor **359** configured to perform the functions recited by the means.
(140) FIG. **14** is a flowchart **1400** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, **402**, **404**, **406**, **504**, **506**, **1006**; the apparatus **1604**). The method may enable the UE to receive SL communication from another UE via resources associated with one or more UL transmissions of an UL-CG.
(141) At **1402**, the UE may receive an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node, such as described in connection with FIG. **10**. For example, at **1022**, the second UE **1006** may receive an UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** from the base station **1002** and/or the first UE **1004**, where the UL-CG configuration **1008** includes a set of periodic resources **1010** for the first UE **1004**. The reception of the indication of the UL-CG configuration may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**.
(142) In one example, the indication is received from the first UE via a second SL transmission or received from the network node via a UE-to-universal mobile telecommunications system (UMTS)

terrestrial radio access network (UE-to-UTRAN) (Uu) link.

(143) In another example, the indication includes a subset of parameters of the UL-CG configuration.

(144) In another example, the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.

(145) In another example, the UE corresponds to a transmitting SL UE and the second UE corresponds to a receiving SL UE.

(146) At **1404**, the UE may monitor for the SL transmission based on the set of periodic resources prior to receiving the SL transmission, such as described in connection with FIG. **10**. For example, at **1026**, the second UE **1006** may monitor for SL transmission based on the UL-CG configuration **1008**. The monitoring of the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**. In one example, monitoring for the SL transmission based on the set of periodic resources includes: monitoring all of the set of periodic resources, monitoring a second subset of the set of periodic resources, or monitoring at least one periodic resource in the set of periodic resources associated with the UL-CG configuration.

(147) At **1406**, the UE may receive a second indication from the first UE indicating the SL transmission, where the SL transmission is received based on the second indication, such as described in connection with FIG. **10**. For example, at **1028**, the second UE **1006** may receive an indication **1012** indicating a sidelink transmission from the first UE **1004**. The reception of the indication of the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**.

(148) At **1408**, the UE may receive a configuration that configures one or more parameters for receiving the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration, such as described in connection with FIG. **10**. For example, the second UE **1006** may receive a configuration that configures one or more parameters for receiving the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration. The reception of the configuration that configures one or more parameters for receiving the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**. In one example, the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or a reception (Rx) power for receiving the SL transmission.

(149) At **1410**, the UE may receive a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration, such as described in connection with FIG. **10**. For example, at **1032**, the second UE **1006** may receive a SL transmission from the first UE **1004** via at least a subset of the set of periodic resources **1010** associated with the UL-CG configuration **1008**. The reception of the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**.

(150) FIG. **15** is a flowchart **1500** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, **402**, **404**, **406**, **504**, **506**, **1006**; the apparatus **1604**). The method may enable the UE (e.g., a first UE) to receive SL communication from another UE (e.g., a second UE) via resources associated with one or more UL transmissions of an UL-CG configured for the another UE.

(151) At **1502**, the UE may receive an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node, such as described in connection with FIG. **10**. For example, at **1022**, the second UE **1006** may receive an UL-CG configuration **1008** or a portion of the UL-CG configuration **1008** from the base station **1002** and/or the first UE **1004**, where the UL-

CG configuration **1008** includes a set of periodic resources **1010** for the first UE **1004**. The reception of the indication of the UL-CG configuration may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**.
(152) At **1510**, the UE may receive a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration, such as described in connection with FIG. **10**. For example, at **1032**, the second UE **1006** may receive a SL transmission from the first UE **1004** via at least a subset of the set of periodic resources **1010** associated with the UL-CG configuration **1008**. The reception of the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**.
(153) In one example, the indication is received from the first UE via a second SL transmission or received from the network node via a Uu link.
(154) In another example, the indication includes a subset of parameters of the UL-CG configuration.
(155) In another example, the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.
(156) In another example, the UE corresponds to a transmitting SL UE and the second UE corresponds to a receiving SL UE.
(157) In another example, the UE may monitor for the SL transmission based on the set of periodic resources prior to receiving the SL transmission, such as described in connection with FIG. **10**. For example, at **1026**, the second UE **1006** may monitor for SL transmission based on the UL-CG configuration **1008**. The monitoring of the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**. In such an example, monitoring for the SL transmission based on the set of periodic resources includes: monitoring all of the set of periodic resources, monitoring a second subset of the set of periodic resources, or monitoring at least one periodic resource in the set of periodic resources associated with the UL-CG configuration.
(158) In another example, the UE may receive a second indication from the first UE indicating the SL transmission, where the SL transmission is received based on the second indication, such as described in connection with FIG. **10**. For example, at **1028**, the second UE **1006** may receive an indication **1012** indicating a sidelink transmission from the first UE **1004**. The reception of the indication of the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**.
(159) In another example, the UE may receive a configuration that configures one or more parameters for receiving the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration, such as described in connection with FIG. **10**. For example, the second UE **1006** may receive a configuration that configures one or more parameters for receiving the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration. The reception of the configuration that configures one or more parameters for receiving the SL transmission may be performed by, e.g., the SL process component **197**, the cellular baseband processor **1624**, and/or the transceiver(s) **1622** of the apparatus **1604** in FIG. **16**. In such an example, the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or a reception (Rx) power for receiving the SL transmission.
(160) FIG. **16** is a diagram **1600** illustrating an example of a hardware implementation for an apparatus **1604**. The apparatus **1604** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **1604** may include a cellular baseband processor **1624** (also referred to as a modem) coupled to one or more transceivers **1622** (e.g., cellular RF transceiver). The cellular baseband processor **1624** may include on-chip memory **1624′**. In some

aspects, the apparatus **1604** may further include one or more subscriber identity modules (SIM) cards **1620** and an application processor **1606** coupled to a secure digital (SD) card **1608** and a screen **1610**. The application processor **1606** may include on-chip memory **1606′**. In some aspects, the apparatus **1604** may further include a Bluetooth module **1612**, a WLAN module **1614**, an SPS module **1616** (e.g., GNSS module), one or more sensor modules **1618** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial measurement unit (IMU), gyroscope, and/or accelerometer(s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other technologies used for positioning), additional memory modules **1626**, a power supply **1630**, and/or a camera **1632**. The Bluetooth module **1612**, the WLAN module **1614**, and the SPS module **1616** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **1612**, the WLAN module **1614**, and the SPS module **1616** may include their own dedicated antennas and/or utilize the antennas **1680** for communication. The cellular baseband processor **1624** communicates through the transceiver(s) **1622** via one or more antennas **1680** with the UE **104** and/or with an RU associated with a network entity **1602**. The cellular baseband processor **1624** and the application processor **1606** may each include a computer-readable medium/memory **1624′**, **1606′**, respectively. The additional memory modules **1626** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory **1624′**, **1606′**, **1626** may be non-transitory. The cellular baseband processor **1624** and the application processor **1606** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor **1624**/application processor **1606**, causes the cellular baseband processor **1624**/application processor **1606** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor **1624**/application processor **1606** when executing software. The cellular baseband processor **1624**/application processor **1606** may be a component of the UE **350** and may include the memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **1604** may be a processor chip (modem and/or application) and include just the cellular baseband processor **1624** and/or the application processor **1606**, and in another configuration, the apparatus **1604** may be the entire UE (e.g., see **350** of FIG. **3**) and include the additional modules of the apparatus **1604**.

(161) As discussed supra, the SL process component **197** is configured to receive an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node. The SL process component **197** may also be configured to receive a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration. The SL process component **197** may be within the cellular baseband processor **1624**, the application processor **1606**, or both the cellular baseband processor **1624** and the application processor **1606**. The SL process component **197** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. As shown, the apparatus **1604** may include a variety of components configured for various functions. In one configuration, the apparatus **1604**, and in particular the cellular baseband processor **1624** and/or the application processor **1606**, includes means for receiving an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node. The apparatus **1604** may further include means for receiving a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration.

(162) In one example, the indication is received from the first UE via a second SL transmission or

received from the network node via a Uu link.
(163) In another example, the indication includes a subset of parameters of the UL-CG configuration.
(164) In another example, the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.
(165) In another example, the UE corresponds to a transmitting SL UE and the second UE corresponds to a receiving SL UE.
(166) In another example, the apparatus **1604** may further include means for monitoring for the SL transmission based on the set of periodic resources prior to receiving the SL transmission. In such an example, the means for monitoring for the SL transmission based on the set of periodic resources includes configuring the apparatus **1604** to monitor all of the set of periodic resources, monitor a second subset of the set of periodic resources, or monitor at least one periodic resource in the set of periodic resources associated with the UL-CG configuration.
(167) In another example, the apparatus **1604** may further include means for receiving a second indication from the first UE indicating the SL transmission, where the SL transmission is received based on the second indication.
(168) In another example, the apparatus **1604** may further include means for receiving a configuration that configures one or more parameters for receiving the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration. In such an example, the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or an Rx power for receiving the SL transmission.
(169) The means may be the SL process component **197** of the apparatus **1604** configured to perform the functions recited by the means. As described supra, the apparatus **1604** may include the TX processor **368**, the RX processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX processor **368**, the RX processor **356**, and/or the controller/processor **359** configured to perform the functions recited by the means.
(170) It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not limited to the specific order or hierarchy presented.
(171) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims. Reference to an element in the singular does not mean "one and only one" unless specifically so stated, but rather "one or more." Terms such as "if," "when," and "while" do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., "when," do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term "some" refers to one or more. Combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" may be A

only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. If a first apparatus receives data from or transmits data to a second apparatus, the data may be received/transmitted directly between the first and second apparatuses, or indirectly between the first and second apparatuses through a set of apparatuses. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. Moreover, nothing disclosed herein is dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words "module," "mechanism," "element," "device," and the like may not be a substitute for the word "means." As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase "means for."

(172) As used herein, the phrase "based on" shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase "based on A" (where "A" may be information, a condition, a factor, or the like) shall be construed as "based at least on A" unless specifically recited differently.

(173) The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation. Aspect 1 is a method of wireless communication at a first UE, including: transmitting an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, where the indication is transmitted to at least one second UE; skipping at least one UL transmission associated with the UL-CG configuration, where the skipped at least one UL transmission corresponds to a subset of the set of periodic resources; and transmitting a SL transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration. Aspect 2 is the method of aspect 1, further including: receiving the UL-CG configuration from the network node, where the UL-CG configuration includes the set of periodic resources. Aspect 3 is the method of aspect 2, where transmitting the indication of the UL-CG configuration includes forwarding the UL-CG configuration received from the network node. Aspect 4 is the method of aspect 2, where the indication includes a subset of parameters of the UL-CG configuration received from the network node. Aspect 5 is the method of any of aspects 1 to 4, further including: transmitting a second indication to the at least one second UE indicating the SL transmission, where the SL transmission is transmitted based on the second indication. Aspect 6 is the method of any of aspects 1 to 5, further including: receiving a configuration for an allowance to skip the at least one UL transmission from the network node, where the at least one UL transmission is skipped based on the configuration for the allowance. Aspect 7 is the method of any of aspects 1 to 6, where the at least one UL transmission is skipped based on a priority order between UL communication and SL communication. Aspect 8 is the method of aspect 7, where a plurality of UL transmissions is skipped based on the SL transmission having a higher priority order than the plurality of UL transmissions. Aspect 9 is the method of any of aspects 1 to 8, further including: determining to skip the at least one UL transmission associated with the UL-CG configuration based on a slot type configured for the at least one UL transmission, where the at least one UL transmission is skipped based on the determination. Aspect 10 is the method of aspect 9, where the slot type includes at least one of: an UL slot, a FL slot, a FD slot, or a HD slot. Aspect 11 is the method of any of aspects 1 to 10, further including: receiving a configuration that configures one or more parameters for transmitting the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration. Aspect 12 is the method of aspect 11, where the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or a Tx power for transmitting the SL transmission. Aspect 13 is the

method of any of aspects 1 to 12, where the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission. Aspect 14 is the method of any of aspects 1 to 13, where the first UE corresponds to a transmitting SL UE and the at least one second UE corresponds to a receiving SL UE. Aspect 15 is an apparatus for wireless communication at a first UE, including: a memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to implement any of aspects 1 to 14. Aspect 16 is the apparatus of aspect 14, further including at least one of a transceiver or an antenna coupled to the at least one processor. Aspect 17 is an apparatus for wireless communication including means for implementing any of aspects 1 to 14. Aspect 18 is a computer-readable medium (e.g., a non-transitory computer-readable medium) storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 1 to 14. Aspect 19 is a method of wireless communication at a second UE, including: receiving an indication of an UL-CG configuration including a set of periodic resources, where the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node; and receiving a SL transmission from the first UE via at least a subset of the set of periodic resources associated with the UL-CG configuration. Aspect 20 is the method of aspect 19, where the indication is received from the first UE via a second SL transmission or received from the network node via a Uu link. Aspect 21 is the method of aspect 19 or 20, further including: monitoring for the SL transmission based on the set of periodic resources prior to receiving the SL transmission. Aspect 22 is the method of aspect 21, where monitoring for the SL transmission based on the set of periodic resources includes: monitoring all of the set of periodic resources, monitoring a second subset of the set of periodic resources, or monitoring at least one periodic resource in the set of periodic resources associated with the UL-CG configuration. Aspect 23 is the method of any of the aspects 19 to 22, where the indication includes a subset of parameters of the UL-CG configuration. Aspect 24 is the method of any of the aspects 19 to 23, further including: receiving a second indication from the first UE indicating the SL transmission, where the SL transmission is received based on the second indication. Aspect 25 is the method of any of the aspects 19 to 24, further including: receiving a configuration that configures one or more parameters for receiving the SL transmission, where the one or more parameters are based on at least one configuration parameter of the UL-CG configuration. Aspect 26 is the method of aspect 25, where the one or more parameters include at least one of: a beam direction, a MCS, a bandwidth, or an Rx power for receiving the SL transmission. Aspect 27 is the method of any of the aspects 19 to 26, where the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission. Aspect 28 is the method of any of the aspects 19 to 27, where the first UE corresponds to a transmitting SL UE and the second UE corresponds to a receiving SL UE. Aspect 29 is an apparatus for wireless communication at a second UE, including: a memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to implement any of aspects 19 to 28. Aspect 30 is the apparatus of aspect 31, further including at least one of a transceiver or an antenna coupled to the at least one processor. Aspect 31 is an apparatus for wireless communication including means for implementing any of aspects 19 to 28. Aspect 32 is a computer-readable medium (e.g., a non-transitory computer-readable medium) storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 19 to 28.

Claims

1. An apparatus for wireless communication at a first user equipment (UE), comprising: at least one memory; and at least one processor coupled to the at least one memory, wherein the at least one processor is configured to: transmit an indication of an uplink (UL)-configured grant (CG) (UL-

CG) configuration including a set of periodic resources, wherein the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, wherein the indication is transmitted to at least one second UE; receive, from the network node, a configuration for an allowance to skip at least one UL transmission associated with the UL-CG configuration; skip the at least one UL transmission associated with the UL-CG configuration based on the configuration, wherein the skipped at least one UL transmission corresponds to a subset of the set of periodic resources; and transmit a sidelink (SL) transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration.

2. The apparatus of claim 1, wherein the at least one processor is configured to: receive the UL-CG configuration from the network node, wherein the UL-CG configuration includes the set of periodic resources.

3. The apparatus of claim 2, wherein to transmit the indication of the UL-CG configuration, the at least one processor is configured to forward the UL-CG configuration received from the network node.

4. The apparatus of claim 2, wherein the indication includes a subset of parameters of the UL-CG configuration received from the network node.

5. The apparatus of claim 1, wherein the at least one processor is configured to: transmit a second indication to the at least one second UE indicating the SL transmission, wherein the SL transmission is transmitted based on the second indication.

6. The apparatus of claim 1, wherein the at least one UL transmission is skipped based on a priority order between UL communication and SL communication.

7. The apparatus of claim 6, wherein a plurality of UL transmissions is skipped based on the SL transmission having a higher priority order than the plurality of UL transmissions.

8. The apparatus of claim 1, wherein the at least one processor is configured to: determine to skip the at least one UL transmission associated with the UL-CG configuration based on a slot type configured for the at least one UL transmission, wherein the at least one UL transmission is skipped based on the determination.

9. The apparatus of claim 8, wherein the slot type includes at least one of: an UL slot, a flexible (FL) slot, a full-duplex (FD) slot, or a half-duplex (HD) slot.

10. The apparatus of claim 1, wherein the at least one processor is configured to: receive a configuration that configures one or more parameters for transmitting the SL transmission, wherein the one or more parameters are based on at least one configuration parameter of the UL-CG configuration.

11. The apparatus of claim 10, wherein the one or more parameters include at least one of: a beam direction, a modulation coding scheme (MCS), a bandwidth, or a transmission (Tx) power for transmitting the SL transmission.

12. The apparatus of claim 1, wherein the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.

13. A method of wireless communication at a first user equipment (UE), comprising: transmitting an indication of an uplink (UL)-configured grant (CG) (UL-CG) configuration including a set of periodic resources, wherein the UL-CG configuration corresponds to at least one allocated UL transmission for the first UE to a network node, wherein the indication is transmitted to at least one second UE; receiving, from the network node, a configuration for an allowance to skip at least one UL transmission associated with the UL-CG configuration; skipping the at least one UL transmission associated with the UL-CG configuration based on the configuration, wherein the skipped at least one UL transmission corresponds to a subset of the set of periodic resources; and transmitting a sidelink (SL) transmission to the at least one second UE using at least the subset of the set of periodic resources corresponding to the skipped at least one UL transmission associated with the UL-CG configuration.

14. The method of claim 13, further comprising: receiving the UL-CG configuration from the network node, wherein the UL-CG configuration includes the set of periodic resources.
15. The method of claim 13, further comprising: transmitting a second indication to the at least one second UE indicating the SL transmission, wherein the SL transmission is transmitted based on the second indication.
16. The method of claim 13, further comprising: determining to skip the at least one UL transmission associated with the UL-CG configuration based on a slot type configured for the at least one UL transmission, wherein the at least one UL transmission is skipped based on the determination.
17. The method of claim 13, further comprising: receiving a configuration that configures one or more parameters for transmitting the SL transmission, wherein the one or more parameters are based on at least one configuration parameter of the UL-CG configuration.
18. The method of claim 17, wherein the one or more parameters include at least one of: a beam direction, a modulation coding scheme (MCS), a bandwidth, or a transmission (Tx) power for transmitting the SL transmission.
19. An apparatus for wireless communication at a second user equipment (UE), comprising: at least one memory; and at least one processor coupled to the at least one memory, wherein the at least one processor is configured to: receive an indication of an uplink (UL)-configured grant (CG) (UL-CG) configuration including a set of periodic resources, wherein the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node; receive a second indication from the first UE indicating a sidelink (SL) transmission using at least a subset of the set of periodic resources associated with the UL-CG configuration; and receive, based on the second indication, the SL transmission from the first UE via at least the subset of the set of periodic resources associated with the UL-CG configuration wherein at least one UL transmission is skipped.
20. The apparatus of claim 19, wherein the indication is received from the first UE via a second SL transmission or received from the network node via a UE-to-universal mobile telecommunications system (UMTS) terrestrial radio access network (UE-to-UTRAN) (Uu) link.
21. The apparatus of claim 19, wherein the at least one processor is configured to: monitor for the SL transmission based on the set of periodic resources prior to receiving the SL transmission.
22. The apparatus of claim 21, wherein to monitoring for the SL transmission based on the set of periodic resources, the at least one processor is configured to: monitor all of the set of periodic resources, monitor a second subset of the set of periodic resources, or monitor at least one periodic resource in the set of periodic resources associated with the UL-CG configuration.
23. The apparatus of claim 19, wherein the indication includes a subset of parameters of the UL-CG configuration.
24. The apparatus of claim 19, wherein the at least one processor is configured to: receive a configuration that configures one or more parameters for receiving the SL transmission, wherein the one or more parameters are based on at least one configuration parameter of the UL-CG configuration.
25. The apparatus of claim 24, wherein the one or more parameters include at least one of: a beam direction, a modulation coding scheme (MCS), a bandwidth, or a reception (Rx) power for receiving the SL transmission.
26. The apparatus of claim 19, wherein the set of periodic resources is associated with a set of time and frequency resources and a periodicity for the at least one allocated UL transmission.
27. A method of wireless communication at a second user equipment (UE), comprising: receiving an indication of an uplink (UL)-configured grant (CG) (UL-CG) configuration including a set of periodic resources, wherein the UL-CG configuration corresponds to at least one allocated UL transmission for a first UE to a network node; receiving a second indication from the first UE indicating a sidelink (SL) transmission using at least a subset of the set of periodic resources

associated with the UL-CG configuration; and receiving, based on the second indication, the SL transmission from the first UE via at least a the subset of the set of periodic resources associated with the UL-CG configuration wherein at least one UL transmission is skipped.